BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD CAMPUS
EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2026

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1. A single-phase full bridge inverter is feeding to a series load of R=20 Ω and L=25 mH. Input dc voltage is 200 V and fundamental frequency of the output is 50 Hz.
    a. If the inverter is operated under square wave modulation (duty ratio of 50%), calculate (i) fundamental RMS of output voltage, (ii) true RMS of output voltage, (iii) THD in the output voltage, (iii) fundamental RMS of the load current, (iv) average input current. Find the LOH in the output voltage.
    b. If the inverter is operated under phase displacement control with $\alpha=120^{o}$, calculate the above parameters.

    (Approximate the true RMS of load current as its fundamental RMS)

2. A single-phase full bridge inverter operating under unipolar SPWM technique is connected to a load of R=50 Ω, and L=100 mH. If the desired fundamental RMS of the output voltage is 230 V ac, 50 Hz, what should be the minimum dc link voltage considering a safety factor of 1.2? What is the modulation index (M_a) in this case? Calculate the RMS load current and power consumed by the load. If the frequency modulation ratio is 55, what is the lowest frequency around which the side bands will appear in the harmonic spectrum? If the modulation index is changed to 0.5 keeping the dc link voltage same, what will be the fundamental RMS of output voltage and current? What will be the load power consumption?

3. A three-phase inverter is operating under 180^{o} conduction mode. It is feeding star connected balanced load, each phase consists of R=5 Ω and L=10 mH. The input dc link voltage is 400 V and the fundamental frequency of the output voltage is 50 Hz. Calculate the fundamental RMS of the output voltage (L-L and phase) and line current. Find the THD of the output voltage (both L-L and phase). What is the LOH in both the cases?

4. A three-phase inverter, operating under SPWM technique ($f_c >> f_1$), is feeding a balanced star connected load of R=20 Ω/ph and X_L=5 Ω/ph (at rated fundamental frequency of 50 Hz). If the rated line to line fundamental voltage is 400 V, what should be the minimum dc link voltage? Calculate the load current RMS and the power consumed by the load. If V/f is kept constant, what should be the modulation index to operate the load at 40 Hz? Calculate the load current RMS and the power consumed by the load. If the third harmonic injection technique is applied, what can be the minimum dc link voltage to operate the load till rated frequency? Is this technique beneficial?